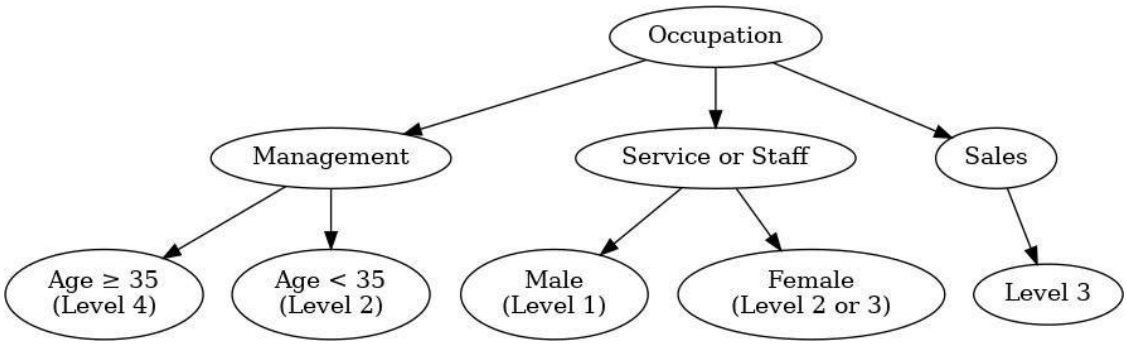
MIS 637 B- DATA ANALYTICS AND MACHINE LEARNING

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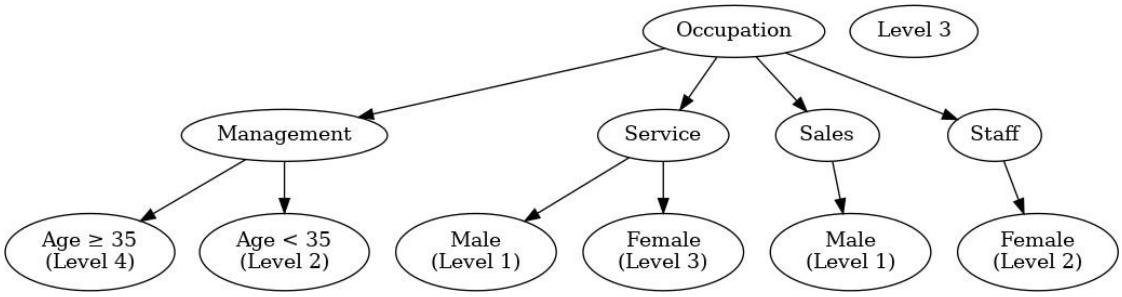
Discretized Salary Levels

Occupation	Gender	Age	Salary	Salary Level
Service	Female	45	$48,000	Level 3
Service	Male	25	$25,000	Level 1
Service	Male	33	$35,000	Level 2
Management	Male	25	$45,000	Level 2
Management	Female	35	$65,000	Level 4
Management	Male	26	$45,000	Level 2
Management	Female	45	$70,000	Level 4
Sales	Female	40	$50,000	Level 3
Sales	Male	30	$40,000	Level 2
Staff	Female	50	$40,000	Level 2
Staff	Male	25	$25,000	Level 1

5. CART Decision Tree



6. C4.5 Decision Tree



7. Comparison of CART and C4.5 Decision Trees

CART (Classification and Regression Trees):

- Uses binary splits, which can make the tree more complex but structured.
- Works well with numerical and categorical data.
- Uses Gini impurity for splitting nodes.

C4.5 Decision Tree:

- Allows multi-way splits, making the tree potentially more compact.
- Uses entropy and information gain for node splitting.
- Handles missing values better than CART.

Comparison:

- CART produces strictly binary trees, while C4.5 can have multi-way splits, reducing depth.
- C4.5 generally provides more interpretable rules but can be computationally intensive.
- Both methods classify the salary levels correctly but may differ in efficiency depending on the dataset.

8. CART Decision Tree Rules

1. If Occupation = Management and Age >= 35, then Salary Level = 4.
2. If Occupation = Management and Age < 35, then Salary Level = 2.
3. If Occupation = Service and Gender = Male, then Salary Level = 1.
4. If Occupation = Service and Gender = Female, then Salary Level = 3.
5. If Occupation = Sales, then Salary Level = 3.
6. If Occupation = Staff and Gender = Male, then Salary Level = 1.
7. If Occupation = Staff and Gender = Female, then Salary Level = 2.

9. C4.5 Decision Tree Rules

8. If Occupation = Management and Age >= 35, then Salary Level = 4.
9. If Occupation = Management and Age < 35, then Salary Level = 2.
10. If Occupation = Service and Gender = Male, then Salary Level = 1.
11. If Occupation = Service and Gender = Female, then Salary Level = 3.
12. If Occupation = Sales, then Salary Level = 3.
13. If Occupation = Staff and Gender = Male, then Salary Level = 1.
14. If Occupation = Staff and Gender = Female, then Salary Level = 2.

10. Comparison of Decision Rules CART Decision Rules:

- Rules are structured with binary conditions, making them easier to implement in models.
- More steps might be needed to reach a classification.

C4.5 Decision Rules:

- Rules can be more concise due to multi-way splits.
- They provide a more direct path to classification but require more computational resources.

Advantages and Disadvantages:

- CART rules are easier to use in automation but might lead to deeper trees.
- C4.5 rules can be more compact but require more sophisticated handling of continuous data and missing values.

Conclusion

Both CART and C4.5 provide effective classification mechanisms for salary prediction. The choice between them depends on the complexity of data, need for interpretability, and computational efficiency. CART's binary structure is useful for stepwise implementation, while C4.5's multi-way splits help streamline classification paths. Depending on the use case, either approach may be preferred.